

The Sports Analytics Controversy in the U.S.

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The Sports Analytics Controversy in the U.S.

Data analytics divides team managers, coaches, athletes and fans in the U.S. In 2014, \$125M was invested in sports data analytics in the U.S and this is expected to increase exponentially to \$4.7B by 2021. Smartphone and social media data, in addition to cloud computing, drive this growth (Tichy, 2016). Though analytics has led some to question the value of human managers and coaches, coaching excellence cannot be entirely reduced to quantitative metrics. Analytics firms, front office employees, sports commentators, athletes, and fans are competing to define the proper limits of analytics in sports.

Review of Research

Most researchers who have investigated the effects of human coach and management practices believe it is an essential part of sports (Jones, 2009). According to Armour (2010), researchers have demonstrated coaches are indispensable to athletes. Jowett (2017) stresses the importance of a coach's relationship with players: "-perceptions of both coach leadership and the coach-athlete relationship predicted variance in team efficacy." Good coaching can motivate players and contribute significantly to their performance as well as the entire sporting experience (Gilchrist & Mallett, 2017). Legg, Newland, and Bigelow (2018) studied youth coaching in basketball, finding coaches who motivate players promote positive youth development. Benjamin Alamar asserted that merging analysis and decision making would be difficult because decision makers do not spend time researching ideas and theories, so they do not see the same vision as analysts. "Decision-makers get used to looking at a particular set of information and unless they are motivated to expand that, it is unlikely they will, no matter how well the analyst makes his case" (Alamar 2013).

Researchers disagree about the merits of sports analytics; all agree analytics are valuable, but they are divided about its reliability and proper applications. Some contend data analytics may supplement or displace human management. Software can track gameplay, helping coaches evaluate plays, scenarios and the players and combinations within (Janke, 2019). Bačić (2012) contends: “qualitative assessment can be automated; assessment diagnostics can be communicated from a machine to human and coaching insights as implicit knowledge can be modeled using connectionist and evolving connectionist approaches.” Some question the ethics of analytics: “The will to win could exceed the limits of what is permitted in fair-play, like the use of prohibited methods” (Vargas-Mendoza, 2018). An example of the misuses of analytics can be seen in the case of the Houston Astros of 2018, who used in-game video technology to predict pitches (Vigdor, 2020). Vargas-Mendoza also suggests analytics may detract from the intrinsic value of sports. According to Miller (2016), sports analytics offers a competitive advantage. He characterizes analytics as a “blend of business savvy, information technology, and modeling techniques.” Jessica Gelman, co-founder and co-chair of the MIT Sloan Sports Analytics Conference says: “Some sports have been quicker to accept than others, but it’s widely accepted at this point that there’s significant value to having analytics to support decision making” (Hunter, 2014).

As a parallel to sports analytics, in a study on the use of automation in planes and its interactions with pilots, Wiener and Curry (1980) argue “there is some reason to believe flight-deck automation may have already passed its optimal point.” They also warn, “It is highly questionable whether total system safety is always enhanced by allocating functions to automatic devices rather than human operators.” The analysis endorses the value of flight automation while

warning of the hazards of excessive reliance on it. In sports analytics, just like flight automation, an optimum is proposed that confirms the value of human judgment in the system.

Analytics Cannot Be Ignored

Sports analytics is here to stay. With the advent of technologies like tracking cameras and Hadoop, and with the steady ‘influx’ of analysts, investors, and ideas across the socio-political-economic landscape, the volume and variety of data only grows, as does their speed of movement. In this context, athletes are newly evaluated while fans become both ‘fanalysts’(fan-analysts) and the targets of corporate surveillance in novel ways (Millington & Millington, 2015). Data hints at the often-strong relationship between belief and performance, specifically, an athlete's belief in a substance, piece of equipment, technique, or even another person, might be a significant factor in their subsequent success (Beedie, 2007). This placebo effect described by Beedie can also be interpreted in the realm of analytics. If players believe in the positive effects of analytics, even if they are doing nothing, players will still see an improvement in ability or performance. To maintain the advantages of analytics, Billy Beane, the creator of Moneyball, stresses long-term trends: “Making assumptions [based] on one winter, or one year? I mean, how many times has the new thing become old” (Sheinin 2018).

Analytics offers the opportunity to study various aspects of different sports previously not observed. According to Jayal et al. (2018), data on the velocities of balls and players offer “a great opportunity for sports analysts to gain critical insights into their team’s performance, which could translate to a competitive edge over rivals.” It is difficult for the eye to glean information from tracking movement and velocities in-game, whereas with the use of high-speed cameras, these measures are accomplished with relative ease. As Jacob Mortensen and Luke Bornn,

analysts from Simon Fraser University, put it, “Analysis of this [player movement] data presents a substantial challenge, due to both the scale of the data, which can consist of more than 100 million rows for a given season, and to the sophisticated methods required to make sense of it” (Mortensen & Bornn, 2019). While the sheer amount of data can be an initial barrier to analysis, the quantity of data is also always a key component in determining the analysis’ effectiveness. With the staggering amount of player movement data available, good analysis is more than possible and solid conclusions are reachable. With the rise of optical tracking data, accurate modeling of player movement has become a competitive advantage in many sports (Mortensen & Bornn, 2019). Nowadays, there are several ways to use data-driven ghosting, a machine learning technique simulating players as ghosts over a large number of plays to quantify, analyze, and compare granular defensive behavior at scale (Le et al., 2017). Using the ghosting software, teams predict what a defensive player should have done instead of what they did, providing an effective teaching moment to players.

Analytics helps with more than just performance on the field, performance on the business side can improve as well. Wolf and Floyd (2013) conclude it is reasonable to expect analytics will help companies because examining data to make decisions often outperforms guesswork and follows the mature literature in strategic planning. These analytics increase revenue through methods such as dynamic pricing, customer relationship management, measuring sponsorships and advanced database marketing (Meenaghan & O’Sullivan, 2013). Analytics can also predict attendance on game days, which is useful for restaurant and supplier expectations (Miller, 2016: 106–109). In a deeper look into the adoption of business analytics in sports organizations, Trolio, Bouchet, Urban, and Sutton state: “The first key finding in sports is that the adoption of analytics by professional sports organizations does convince managers their

revenues are growing. A second finding is these revenue increases are real and significant” (2015).

Analytics Improve Player Safety

The use of data analytics helps prevent injury and reduces the loss of value attributed to a team when a player is injured. Metrics have appeared, such as the acute:chronic workload ratio, defined as the cumulative workload of the past seven days divided by the past 28 days, to help predict injury risk (Hulin et al. 2014). Other researchers have even improved the accuracy of these indicators by adding survival models to control possible confounding effects, such as schedules (Bornn, Ward, Norman, 2019). These metrics judge a player’s workload to determine if they are on the precipice of injury. Models are designed to show, “...by resting the top 20% of high-risk scores at any given day there is a potential to prevent 60% of all injuries” (Talukder et al., 2016). By using these models and metrics, “the direct consequence of reducing the number of player injuries incurred over the course of a season would mitigate the financial losses incurred by organizations” (Talukder et al., 2016).

Using sensors to monitor an athlete’s exertion provides a means of load management, enabling athletes to perform at a higher level. Phillip Coles, a member of the San Antonio Spurs medical staff asserts, “there is a maximum level of load that each player will be able to tolerate, at any given point in time. If they are pushed beyond that level, they will break down” (Coles, 2017). Isaiah Kacyvenski, the head of a leading wearable technology firm MC10’s sports department, claims athletes must “get away from feeling our way through the day and put hard metrics” into an athlete’s daily routine “and guide us through our day” (Springer, 2014). By using sensors, a team can quantitatively assess a player’s health, as opposed to using qualitative

measures, a more inaccurate indicator of healthiness. Then, using the parameters monitored by wearables, analysts can, “detect position-specific patterns in movement, design more efficient sports-specific training programs for performance optimization, and screen for potential causes of injury” (Li et al., 2015).

Limiting the Technological Takeover

While analytics offer highly technical and data-driven perspectives, it is important to maintain a healthy level of skepticism. This skepticism allows players and coaches to maintain, to some degree, their routines and feel for the game. Draymond Green, an NBA All-Star, states, “Professionally, I play completely off of feel. I hear people discussing my game in terms of all these advanced numbers. I have no part of it. Even paying attention to it, from a playing standpoint, would make me robotic and undermine my game” (Wilbon, 2016). Human beings are creatures of routine, and it is important for the players to maintain some degree of autonomy. This athletic independence allows them to perform how they feel most comfortable as a complete analytics takeover would result in many unhappy athletes. As Bryan Curtis, a sports writer, puts it, “If you tell a player he’s lousy and have the numbers to back it up, the player turns his back on you. If you tell a player he’s wonderful and have the numbers to back it up, the player cracks a smile and then turns his back on you” (Curtis, 2015). By limiting the extent to which teams rely on analytics, players can stay freer, happier, and on top of their game.

From an entertainment perspective, sports analytics can be viewed as monotonous and robotic. To maintain a positive fan experience, you must limit their exposure and public use. As former baseball player Jayson Werth puts it, “We’re creating something that’s not fun to watch. It’s boring. You’re turning players into robots” (Leopold, 2018). Well-known sportswriter

Michael Wilbon argues, “The greater the dependence on the numbers, the more challenged people are to tell (or understand) the narrative without them” (Wilbon, 2016). With the increase in the amount of data talk, it can be easy to get lost in the numbers and lose some of the intrinsic value of watching sports. As Larry Irvine, a Stanford lawyer, put it, “Sports is emotional. And analytics represent the absence of emotion, the antithesis. Nobody gets into sports to be dispassionate” (Wilbon, 2016). Without the emotional connection to sports, there would be no fandom and no one to care.

Tracing the origins of the data is a necessary step to proving the integrity of the information and results acquired. Stan Van Gundy, a former NBA Head Coach and now analyst, asks whether the people recording the data are knowledgeable enough to understand the intricacies of the sport and the acute differences in certain events. He added, “one of the problems is, there are a lot of people in a lot of organizations who don’t know the game, who all they know is analytics and as a result, that’s what they rely on. There’s no substitute for watching film over and over and over again, and the only numbers I trust are the ones that my people believe” (Pollakoff, 2014). Charles Barkley, a former NBA player and now commentator, takes it even further on those who study analytics, “They’re a bunch of guys who ain’t never played the game” (Boren, 2015). Dr. David Schrader, a national speaker on analytics, believes the industry is behind in creating a centralized athletic database, which raises some concerns: “If an athlete doesn’t play well, it’s almost impossible to backtrack and easily ‘see’ what the contributing factors might have been because the correlations across these various data sets can’t be found” (Richter, 2019). For best use, analytics need to be centralized, interpreted, and evaluated by those familiar with the sport; these tactics provide the best means to get players and coaches to buy into the benefits analytics provide.

Measurable Benefits of Coaching

Coaching goes beyond X's and O's, there are measurable benefits to good coaching in sports. After developing a method for estimating coaching effects, researchers Berry and Fowler, had this to say: "Our estimates generally imply coaches explain about 20-30 percent of the variation in a team's success, although coaching effects vary notably across settings and across various outcomes" (Berry & Fowler, 2019). Berri et al. (2009) employed coach-dummy variables to explain variation in the performance of NBA players. Of the 62 coaches studied, 14 had a statistically significant individual impact. The percentage of coaches making a significant impact is not large, but it does show having the right person at the helm can lead to better results. Brian Goff (2013), used a hierarchical model showing that 8.5% of the variance in team win rates is attributable to managers in MLB, and 21% of the variation is attributable to coaches in the NFL. Numbers can capture a coach's influence but they cannot replace it.

Good coaches provide an environment for the players to succeed. According to Ames (1992), coaches are responsible for creating the motivational climate of the team; this climate influences the goals and enjoyment of the athletes. This motivational climate contributes to a player's orientation, be it ego or task-oriented, the difference lies in how they interpret losses and the impact on their motivation (Duda & Whitehead, 2016). Pensgaard and Roberts (2002, 1992) found an individual's personal goal perspective and perceived motivational climate is significant. Mageau and Vallerand (2003) contend, when coaches support athletes' autonomy, the athletes have better performances, increased persistence, increased self-determined motivation, and enhanced psychological well-being. Coaches must provide the player with the right environment without the impact of analytics on its effectiveness.

Coaches provide a psychological edge. A key attribute of success is a positive coach-athlete relationship; a coach can provide three psychological constructs – closeness, commitment, and complementarity. Jowett and Ntoumanis (2004) assert by attaining these constructs, you can expect optimal performance. To Horn (2008), athletes perceive coach behavior in a personalized way and it is these subjective perceptions that indirectly affect athletes' self-perceptions, beliefs, and attitudes. In turn, this impacts motivation levels and ultimately their performance and behavior. Wilson, Cushion, and Stephens (2006) reported that coaches' expectancies of athletes have the potential to influence the subsequent behavior and performance of sports performers. A good coach can inspire players to perform better, without numbers.

Compromise Is Key

For optimal use, analytics must be used in harmony with the human element of coaching. Past athletic stars, such as Kristine Lilly, are concerned over receiving data in real-time: "There should be a balance with what you share with the athlete and when you share it. You don't want to say to an athlete, 'Oh, you're not working hard because your heart rate is this.'" Springer (2014) contends, a good coach knows their players and will know best how to approach situations such as these. Phillip Coles, member of the San Antonio Spurs Performance and Medicine department, on coaches decision making for load management: "It should be realized, most of the actual content in terms of in-season player load will still be coach driven. So, while the high-performance staff may have expertise in terms of how much load different players can tolerate, if they are not working hand in hand with the coaching staff, they are unlikely to be successful in preventing injuries" (Coles, 2017). Talukder et al. (2016) proposes, "With our

model, there is now a data-driven way to strategically rest players throughout the year. The head coach or general manager can combine the probability with the importance of the opponent to make a more data-driven decision. A coach can start to manage minutes played by the NBA player from the start of the season in a more effective way,” further improving the analytic blend. According to Kuehn (2016), analytics can now numerically, “-identify which players are good and bad teammates, and also separate how much of a player’s statistical output is due to their individual skills and how much is due to the team they play on.” Jalen Rose, former NBA player and now commentator: “I think analytics should be a tool in the box, not the actual toolbox” (Chotiner, 2019). The main criterion that distinguishes analytics from more routine data analysis activities, is the integrated use of computerized data analysis techniques along with human expertise (Keim et al., 2010: 149).

Coaching and analytics have important aspects the other does not. According to Terry Stotts an NBA head coach: “The one thing about analytics is they’re cold numbers, and they don’t take into account the dynamics of the team” (Sunnergren, 2014). NBA player Shawn Livingston adds to this sentiment: “I think the analytics are really overrated when it comes to putting a team together. What statistical analysis is going to tell you whether a guy is a good teammate, whether he can lead, whether he has his teammate’s back? Don’t you have to see and figure that stuff out independent of any numbers” (Wilbon 2016)? Analytics, however, can quantitatively examine and compare patterns of individual player movements: “When a coach wants to look at a new play or movement, the entire process [a film analyst watching hundreds of hours of game footage] must be repeated. This approach is neither efficient (it takes hundreds of hours) nor comprehensive (film analysts don’t retrieve all examples of a certain movement)”

(Nistala & Guttag, 2019). Miller (2016) contends, in some ways, computers can do well, while in others, human experts can do better.

Conclusion

The integration of data analytics into sports was not a sudden development, but one developed over time, lending to further growth in the future. Although current work has achieved reliable performance, there are many other features and analyses that can further complement previous research. First, increasing the breadth of information collected and made available could further improve the accuracy of analytics. For example, statistics on specific play frequency and effort required could unveil more information on susceptibility to injury. Second, by using real-time frameworks, models can scale and generate probabilities on a nightly basis, providing instant feedback. Future work can also expand business benefits investing more in improving the fan experience. By providing data-centric processes, analysts add support, informing coaches and management of ideal actions and measures to improve team success.

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